Two-Dimensional Signal Processing: Discrete Space

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Key Concepts

1) Two-dimensional (2D) discrete-space signal processing involves 2D arrays of numbers called pixels.

2) 2D discrete-space signal processing is a natural extension of one-dimensional discrete-time signal processing.

    a) A 2D linear shift invariant system is completely described by its impulse response, which is called a point-spread function.

    b) The output of a 2D LSI system is the convolution of the input and the system's point-spread function.

c) 2D convolution involves

i) Flip the point-spread function horizontally and vertically.

ii) Shift the flipped point-spread function to the desired output location.

iii) Multiply the flipped and shifted point-spread function with the input signal and sum all values in the product.

3) A point-spread function with all values the same sign will blur the image by averaging pixels.

4) If the point-spread function elements in a row (column) have the same sign, then the image will be blurred across rows (columns).

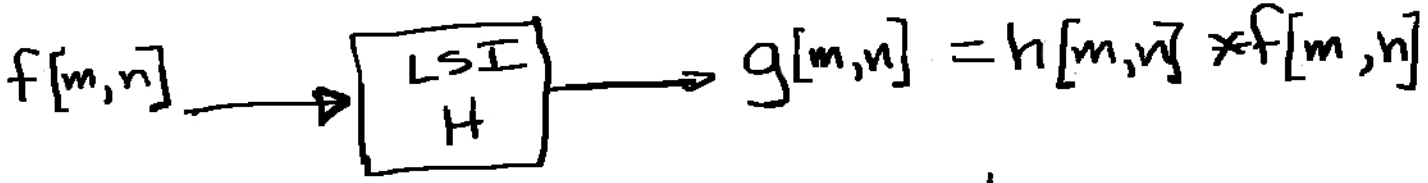
5) If the elements in a row (column) of the point-spread function have opposite signs, then differences within rows (columns) will be accentuated.

2-D Signal Processing: Discrete Space

Discrete space: 2-D array of numbers, "pixels"



Linear Shift Invariant Systems



$h[m,n] = H\{\delta[m,n]\}$ ← impulse response / point spread function

$$g[m,n] = h[m,n] * f[m,n]$$

$$= \sum_{k=-\infty}^{\infty} \sum_{l=-\infty}^{\infty} h[m-k, n-l]\, f[k,l]$$

2-D Convolution – Let $f[m,n] = 0 \quad n<0, n\geq N$

$$g[m,n] = \sum_{k=0}^{N-1} \sum_{\ell=0}^{N-1} h[m-k, n-\ell]\, f[k,\ell]$$

1) Flip $h[k,\ell]$ horizontally and vertically to get $h[-k,-\ell]$
2) Shift $h[-k,-\ell]$ by $[m,n] \Rightarrow h[0,0]$ is at $[m,n]$
3) Form product $h[m-k,n-\ell]f[k,\ell]$ and sum all values

Example:

$$\underline{h} = \begin{bmatrix} h[0,0] & h[0,1] \\ h[1,0] & h[1,1] \end{bmatrix}$$

$(h[m,n]=0 \quad m,n>1 \quad m,n<0)$

$$\underline{f} = \begin{bmatrix} f[0,0] & \cdots & f[0,N-1] \\ \vdots & & \vdots \\ f[N-1,0] & \cdots & f[N-1,N-1] \end{bmatrix}$$

$\underline{g} = \underline{h} * \underline{f}$

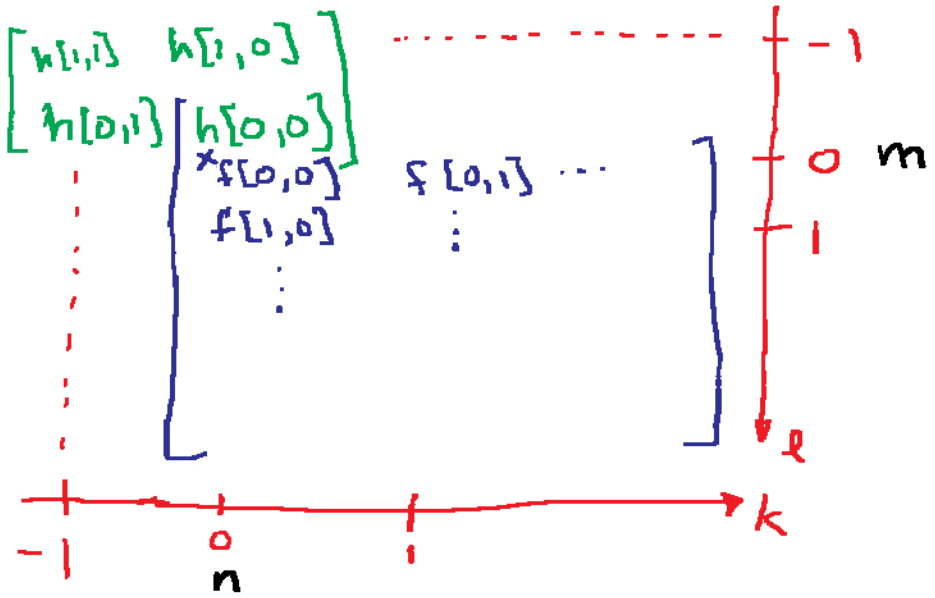
Steps 1 + 2: $h[m-k,n-\ell]$

$$\begin{bmatrix} h[1,1] & h[1,0] \\ h[0,1] & h[0,0] \end{bmatrix} \leftarrow m$$

location n ↑ loc.

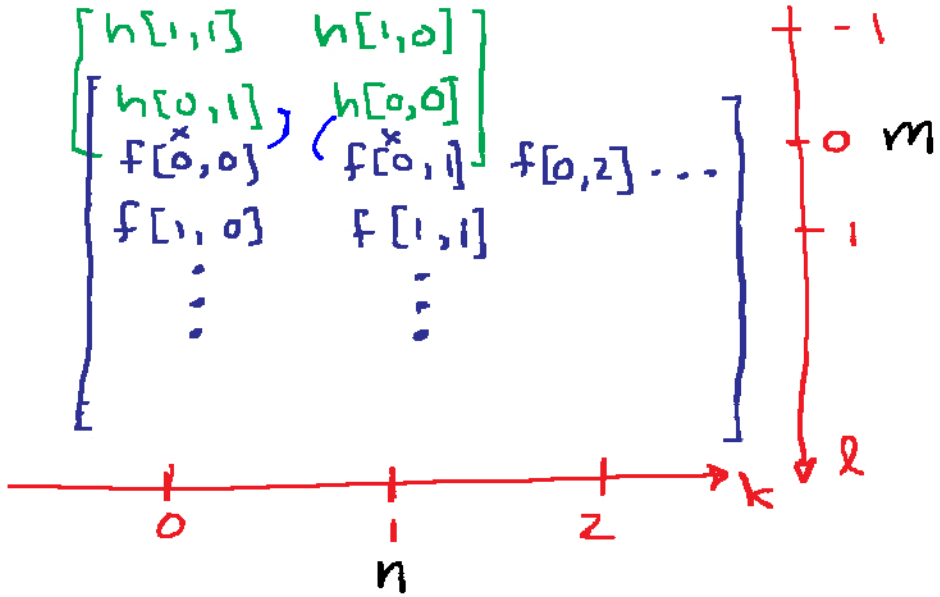
Step 3 : multiply and Sum

$g[0,0]$: $m=0, n=0$



$g[0,0] = h[0,0]\,f[0,0]$

$g[0,1]$: $m=0, n=1$



$g[0,1] = h[0,1]\,f[0,0] + h[0,0]\,f[0,1]$

In general —

$$g[m,n] = h[0,0]\,f[m,n] + h[0,1]\,f[m,n-1] + h[1,0]\,f[m-1,n] + h[1,1]\,f[m-1,n-1]$$

- cols of $\underline{h}$ filter cols of $\underline{f}$ ✓
- rows of $\underline{h}$ filter rows of $\underline{f}$ ✓

Examples for $\underline{h}$

1) $\underline{h} = \frac{1}{4}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

blurs image - each pixel replaced by average of 4 pixels

2) $\underline{h} = \frac{1}{2}\begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}$

blurs vertically
accentuates differences horizontally

3) $\underline{h} = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$

accentuates differences vertically
blurs horizontally

4) $\underline{h} = \frac{1}{9}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

blurs image - more than 1)